16 Prove algebraically that $0.2\dot{5}\dot{6}$ can be written as $\frac{127}{495}$

Let $x = 0.2565656...$

$$1000x = 256.565656...$$

$$10x = 2.565656...$$

Subtract: $990x = 254$

$$x = \frac{254}{990} = \frac{127}{495}$$

Therefore the recurring decimal equals $\frac{127}{495}$

1 mark
Show the repeating block clearly

1 mark
Matching decimal tails cancel

1 mark
Divide numerator and denominator by 2

(Total for Question 16 is 3 marks)